Detecting Face-Swap Deepfakes by Verifying Skeletal Gait Dynamics

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***Abstract*—Face-swap deepfake generators synthesise a face and composite it back into an otherwise unmodified frame. The body they leave behind still walks the way the source person walks. We exploit this asymmetry directly: rather than searching for artefacts inside the manipulated region, we verify the claimed identity against the one signal the generator never touches. Given a video and a claimed identity, our system extracts a 78-dimensional skeletal gait descriptor per frame—hip-centred three-dimensional coordinates, six joint flexion angles and frame-to-frame velocities over 12 biomechanically selected MediaPipe landmarks—and classifies the per-timestep discrepancy between the observed sequence and the claimed identity's enrolled signature using a temporal convolutional verification network. We evaluate on GaitDeepfake-13, a 13-subject corpus of 66 walking videos expanded to 1,056 by a 16× augmentation pipeline, under strict leave-one-subject-out cross-validation in which every video of the held-out subject, augmented or not, is withheld from training. Over 2,240 pooled verification pairs the system attains a pooled ROC-AUC of 94.95% (per-fold $95.10\%\pm3.08\%$), accuracy $87.04\% \pm 3.65\%$, F1 $87.12\% \pm 3.77\%$ and an equal error rate of $12.27\% \pm 3.80\%$. Gradient-based attribution shows the decision resting on shoulder countermotion and distal heel-strike and toe-off kinematics, and shows joint angles carrying $1.94\times$ their proportional share of attribution despite occupying only 6 of 78 input dimensions. On three FaceFusion-generated face-swap clips, the system rejected the claimed identity in all three cases and matched the true body source with similarity above 0.9998, an existence check consistent with the core hypothesis though too small to be a powered claim. An ablation over 13 folds and 3 seeds finds that wiring a CNN, BiLSTM or Transformer encoder onto the decision path costs 2.5 to 5.9 ROC-AUC points against this minimal design (all $p < 10^{-3}$, paired), while removing the raw per-timestep comparison collapses performance to chance—so the simplicity of the verification network is a result, not a concession. We position the work as occupying an under-explored niche rather than an empty one, and delimit it against the behavioural-biometric and whole-body motion forensics literature.**

***Index Terms*—Deepfake detection, gait analysis, behavioural biometrics, pose estimation, identity verification, skeletal keypoints, media forensics, explainable AI.**

I. Introduction

THE dominant paradigm in video deepfake detection is to look for evidence of manipulation inside the manipulated region. Detectors learn blending boundaries, frequency-domain irregularities, warping residues and temporal flicker in and around the face crop [1]–[5]. The paradigm is productive, and it is also structurally fragile: it places the detector in direct opposition to the generator's objective. Every improvement in synthesis quality erodes precisely the signal the detector depends on, which is why cross-dataset generalisation remains the field's most stubborn problem—methods that approach saturation within a training corpus lose substantial ground on unseen manipulations and unseen generators [1], [3], [6].

This paper takes the opposite stance. Instead of interrogating the region the adversary controls, we interrogate a region the adversary does not modify at all.

The observation is mechanical rather than statistical. Contemporary face-swap pipelines—FaceFusion with InSwapper [7], [8], SimSwap [9] and their descendants—detect a face, synthesise a replacement, and composite it back through a mask confined to the facial region. The torso, limbs, and background are carried through from the source footage, modulo recompression. Consequently a face-swapped video of person B built on footage of person A contains B's face and A's *gait*. Human locomotion is constrained by skeletal proportion, mass distribution, and individual motor control, and is a well-established biometric in its own right [10]–[14]. The forgery therefore carries, in plain sight, an identity signal that contradicts its own claim.

A. Problem Formulation

We treat detection as identity verification rather than binary artefact classification. Given a video V and a claimed identity I_c drawn from an enrolled gallery, the system computes

$$f(V, I_c) \to \begin{cases} \textsc{authentic} & \text{if the gait in } V \text{ matches } I_c, \\ \textsc{deepfake} & \text{otherwise.} \end{cases} \tag{1}$$

This framing carries a consequence worth stating plainly at the outset: the system detects an *identity inconsistency*, not the presence of synthesis. It requires an enrolled reference for the claimed identity and it will not flag a wholly synthetic person with no claimed identity. Section VII returns to what this rules in and out.

B. Contributions

1) **A gait-based verification formulation of face-swap detection**, with a threat model in which the discriminative signal lies structurally outside the generator's output support rather than inside it (Section III).
2) **A difference-based temporal verification network** that classifies the per-timestep discrepancy between an

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The GaitDeepfake-13 dataset supporting this work is available on IEEE DataPort (DOI: 10.21227/ngh5-b637) and the evaluation code at https://github.com/Arhaan-P/DeepFake-Detection.

observed gait sequence and an enrolled signature. Operating on explicit sequence differences rather than on distances between separately pooled embeddings avoids the representation collapse we encountered with a Siamese formulation on a corpus of this size (Section III-E).

3) **A subject-disjoint evaluation** over 13 leave-one-subject-out folds and 2,240 verification pairs, reporting per-fold dispersion and two distinct operating points rather than a single aggregate figure (Sections IV–V).
4) **Gradient-based attribution over the skeleton**, which finds the model attending to biomechanically sensible structures, and which quantifies the disproportionate value of the six joint-angle dimensions (Section VI).
5) **An ablation that justifies the architecture by refuting the alternative.** Across 13 folds and 3 seeds, wiring the CNN, BiLSTM or Transformer stack onto the decision path costs 2.5 to 5.9 ROC-AUC points against the deployed model (all $p < 10^{-3}$, paired), and removing the raw per-timestep comparison collapses performance to chance. The minimal difference network is not a simplification we settled for; it is the best of the configurations tested (Section V-D).

II. Related Work

A. Face-Based Deepfake Detection

Detection of facial manipulation has converged on spatial artefact modelling, temporal coherence modelling, or a hybrid. Multi-attentional detection [2] treats forgery detection as fine-grained classification, aggregating evidence from multiple attended regions. RECCE [1] learns a reconstruction of genuine faces and uses reconstruction error as the forgery cue, explicitly targeting generalisation; the paper's own framing is instructive, observing that earlier methods degrade towards approximately 60% AUC on unseen data, and reporting roughly 70.9% cross-dataset AUC on Celeb-DF-v2 for its own approach. FTCN [3] removes spatial convolution capacity to force reliance on temporal incoherence, reaching roughly 86–89% AUC on Celeb-DF as a cross-dataset result. AltFreezing [4] alternately freezes spatial and temporal weights in a 3D ConvNet to balance both artefact families, reporting 89.5% cross-dataset AUC on Celeb-DF. FSBI [5] synthesises training forgeries via frequency-enhanced self-blending.

The pattern across this literature is that in-corpus performance is close to saturated while cross-corpus performance is not, and the gap is generator-dependent. Our approach is complementary rather than competitive: it does not examine the facial region at all, and so its failure modes are uncorrelated with those of every method in this subsection.

B. Skeleton-Based Gait Recognition

Model-based gait recognition operates on pose sequences rather than silhouettes, which confers robustness to clothing and carried objects. GaitGraph [10] applies graph convolution to skeleton sequences. GaitPT [11] shows a hierarchical pose transformer reaching 82.6% average accuracy on CASIA-B from skeletons alone. GaitFormer [12] reports 92.5% on CASIA-B after multi-task pretraining on a large automatically-annotated corpus. GPGait [15] targets cross-domain generalisation from pose. Liao *et al.* [16] incorporate human prior knowledge into a model-based recogniser.

This body of work establishes that skeletal kinematics carry sufficient identity information for recognition at scale. We inherit that premise and redirect it: our task is not to recognise an unknown walker among many, but to adjudicate a specific claimed identity, which is a verification problem with its own metric conventions [17].

C. Behavioural and Whole-Body Forgery Detection

Behavioural biometrics entered deepfake detection with Agarwal *et al.* [18], who modelled correlations among facial action units and head motion to build per-individual behavioural profiles, later combining static facial and temporal behavioural biometrics [19] and subsequently extending beyond the face to gestural and vocal mannerisms [20]. More recently, attention has moved to the whole body. Human Action CLIPs [21] detects AI-generated human motion. TT-DF [22] provides a large-scale benchmark for human body forgery detection exploiting pose and motion inconsistency. Chapariniya *et al.* [23] identify people from whole-body conversational keypoints, framed explicitly against face-reenactment threats. Closest to the present work, DeAndres-Tame *et al.* [24] evaluate whether generative human-animation models preserve gait identity cues, and find that they do not—an evaluation study rather than a deployed detector, but one that independently supports the premise this paper builds on.

D. Positioning

We state our position precisely, because a looser claim would not survive contact with the literature above. Behavioural biometrics have been applied to deepfake detection since 2019 [18], and whole-body pose and motion cues have been applied since at least 2024–2025 [21]–[23]. What remains under-explored, to our knowledge, is the use of *locomotion gait-cycle skeletal dynamics as the primary discriminative feature* for verifying a claimed identity in a face-swapped video, evaluated under subject-disjoint protocol against generator-produced forgeries. The nearest work [24] measures gait fidelity in generated animation without building a detector; the nearest detectors [21], [22] target whole-body synthesis rather than face-swap, where the body is authentic and only the claim is false. We therefore describe this as an under-explored niche, not a vacant one, and Table XIII is offered as positioning rather than as a leaderboard.

III. Proposed Method

Fig. 1 gives an overview of the end-to-end verification pipeline described in this section, from raw video to verdict.

A. Threat Model

We assume an adversary who produces a face-swapped video: a genuine recording of a source subject with a target subject's face composited into the facial region, presented

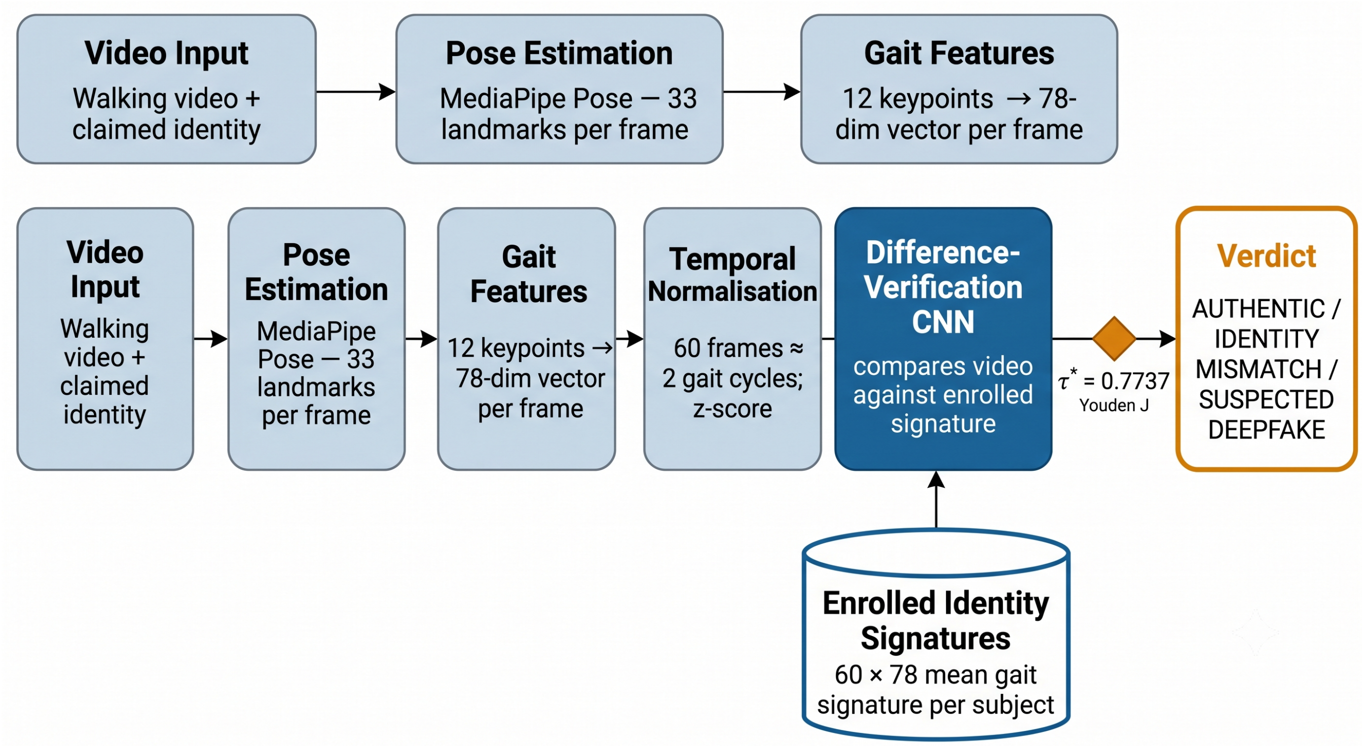

Fig. 1. End-to-end verification pipeline. A walking video and a claimed identity enter at the left; MediaPipe Pose yields 33 landmarks per frame, of which 12 gait-relevant keypoints are retained and expanded into a 78-dimensional per-frame descriptor. Sequences are resampled to $T = 60$ frames and z-score normalised using statistics computed on the training split only. The verification network compares the resulting sequence against the claimed identity's enrolled signature and emits a calibrated score, which the Youden-optimal threshold $\tau^* = 0.7737$ converts into a verdict. Note that the enrolled signature enters the network as a second input sequence, not as a stored embedding.

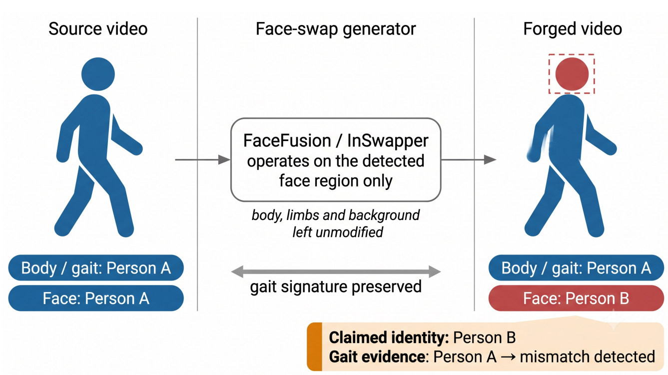

Fig. 2. The asymmetry the method exploits. A face-swap generator detects a face, synthesises a replacement, and composites it back through a face-confined mask; the torso, limbs and background of the source footage pass through essentially unmodified. A forgery claiming to depict person B therefore carries person A's gait. Because the generator never models locomotion, this inconsistency is not an artefact that improved synthesis quality will remove—it is a structural property of face-region-only manipulation.

under the target's identity. The adversary may use any face-swap model and any face-enhancement post-process, and may recompress the result. The adversary does *not* re-synthesise or retime body motion.

Under this model the detector's task reduces to Eq. 1, and its robustness has an unusual character: it is invariant to the quality of the face swap, because it never observes the swapped region. Fig. 2 illustrates the mechanism. Section VII discusses adversaries who violate the final assumption.

TABLE I
COMPOSITION OF THE GAITDEEPFAKE-13 CORPUS

Property	Value
Subjects (enrolled identities)	13
Original recordings	66 (3–5 per subject)
After 16× augmentation	1,056
Viewpoints	Frontal (F), side profile (S)
Frame rate / resolution	30 fps / 720p–1080p
Sequence length	$T = 60$ frames ($\approx$2 s)
Per-frame feature dimension	78
Face-swapped test clips	3 (FaceFusion / InSwapper)
LOOCV verification pairs	2,240 (1,120 genuine)

B. Dataset

Evaluation uses GaitDeepfake-13 [25], collected for this work. Thirteen subjects were recorded walking at natural pace in controlled indoor conditions, from frontal (F) and side (S) viewpoints, yielding 66 original recordings at 30 fps and 720p–1080p. Subjects were students aged approximately 19–25 of both sexes. Table I summarises the corpus.

1) Augmentation: Each recording was expanded 16× ($66 \times 16 = 1{,}056$) by the transformations in Table II. The design constraint was that no transformation may alter the underlying skeletal trajectory. Photometric operations (blur, brightness, contrast, colour jitter, grayscale, noise) change pixels but not the pose MediaPipe recovers from them; geometric rotation

TABLE II
AUGMENTATION OPERATIONS (16× EXPANSION)

#	**Operation**	**Purpose**
1	Original	Baseline
2	Horizontal flip	Bilateral symmetry
3	Gaussian blur (kernel 3–5)	Codec / motion artefacts
4	Brightness -0.2 to -0.3	Low light
5	Brightness $+0.2$ to $+0.3$	Overexposure
6	Contrast $+0.2$ to $+0.3$	High-contrast scenes
7	Colour jitter	Colour invariance
8	Combined multi-augment	Compound robustness
9	Grayscale	Colour invariance
10	Rotation left 5–10°	Camera tilt
11	Rotation right 5–10°	Camera tilt
12	Speed 0.8×	Slow gait
13	Speed 1.2×	Fast gait
14	Temporal reversal	Gait time symmetry
15	Zoom (crop and resize)	Framing variation
16	Gaussian noise injection	Sensor noise

is capped at 10° to avoid anatomically implausible configurations; temporal operations exploit the approximate time-reversal symmetry and natural speed variability of the gait cycle [13], [26].

We note the limitation this implies. Augmentation multiplies the number of training sequences but not the number of *gaits*: the effective sample size for a subject-level generalisation claim remains 13, and Section VII-C treats this accordingly.

C. Pose Estimation and Feature Extraction

1) Landmark Selection: Pose is estimated with MediaPipe Pose Landmarker [27] (`pose_landmarker_lite`, FP16, per-frame IMAGE mode, detection and tracking confidence 0.5), producing 33 three-dimensional landmarks per frame. Of these we retain the 12 listed in Table III and shown in Fig. 3. The selection is driven by gait biomechanics [13], [14], [28]: the lower-limb chain from hip through knee, ankle, heel and foot index spans the events that define the gait cycle—heel strike, stance, toe-off—while the shoulders capture the trunk countermotion that balances angular momentum during swing. Facial, elbow and wrist landmarks are discarded; the facial ones are discarded on principle, since they lie inside the region the adversary controls.

Video-based pose estimation is accurate enough for this purpose. Stenum *et al.* [29] validated markerless video pose estimation against a Vicon optical motion-capture system and reported mean absolute errors of 4.0°, 5.6° and 7.4° for sagittal-plane hip, knee and ankle angles respectively. We cite that result for the general claim it supports—that video pose estimation agrees with laboratory motion capture to within a few degrees—and note that it evaluated OpenPose rather than MediaPipe, so it is corroborating rather than direct evidence for our particular backend.

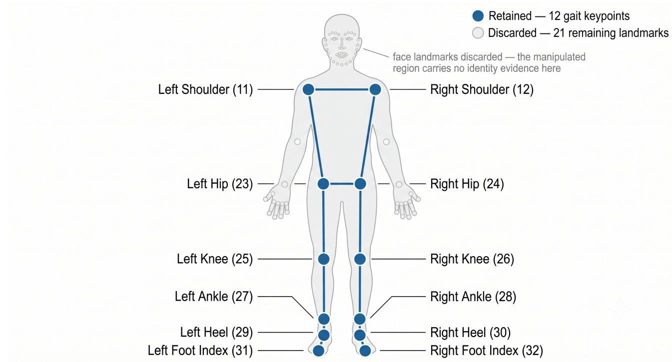

Fig. 3. The 12 gait keypoints retained from MediaPipe's 33 landmarks, shown on the kinematic chain used to compute joint angles. Retained landmarks are the bilateral shoulders, hips, knees, ankles, heels and foot indices; the discarded landmarks—face, elbows, wrists and hands—are drawn de-emphasised. Discarding the facial cluster is a design commitment rather than an economy: it is the region a face-swap adversary controls, and admitting it would reintroduce exactly the dependency this method exists to avoid.

TABLE III
THE 12 RETAINED GAIT LANDMARKS

Idx	**Landmark**	**MP ID**	**Gait role**
0	Left shoulder	11	Trunk countermotion
1	Right shoulder	12	Trunk countermotion
2	Left hip	23	Pelvic motion, CoM
3	Right hip	24	Pelvic motion, CoM
4	Left knee	25	Flexion kinematics
5	Right knee	26	Flexion kinematics
6	Left ankle	27	Foot placement, stride
7	Right ankle	28	Foot placement, stride
8	Left heel	29	Heel strike
9	Right heel	30	Heel strike
10	Left foot index	31	Toe-off
11	Right foot index	32	Toe-off

2) The 78-Dimensional Descriptor: For frame t the descriptor concatenates three blocks,

$$\mathbf{f}_t = [\,\mathbf{c}_t \,\|\, \boldsymbol{\theta}_t \,\|\, \mathbf{v}_t\,] \in \mathbb{R}^{78}, \tag{2}$$

composed as set out in Table IV.

Normalised coordinates $\mathbf{c}_t \in \mathbb{R}^{36}$ are the 12 landmark positions re-expressed relative to the pelvis centre,

$$\hat{\mathbf{p}}_i^{(t)} = \mathbf{p}_i^{(t)} - \tfrac{1}{2}\left(\mathbf{p}_{\text{L\_Hip}}^{(t)} + \mathbf{p}_{\text{R\_Hip}}^{(t)}\right), \quad i = 1, \ldots, 12, \tag{3}$$

which removes translation and much of the scale variation induced by camera distance [10], [11].

Joint angles $\boldsymbol{\theta}_t \in \mathbb{R}^6$ are the three-point flexion angles at both knees, hips and ankles,

$$\theta_{abc} = \arccos\left(\frac{(\mathbf{p}_a - \mathbf{p}_b)\cdot(\mathbf{p}_c - \mathbf{p}_b)}{\|\mathbf{p}_a - \mathbf{p}_b\|\,\|\mathbf{p}_c - \mathbf{p}_b\|}\right), \tag{4}$$

evaluated as $\theta(\text{hip}, \text{knee}, \text{ankle})$, $\theta(\text{shoulder}, \text{hip}, \text{knee})$ and $\theta(\text{knee}, \text{ankle}, \text{foot})$ bilaterally. Angles are scale-free by construction and encode biomechanical constraint directly [13], [28].

Velocities $\mathbf{v}_t = \mathbf{c}_t - \mathbf{c}_{t-1} \in \mathbb{R}^{36}$ are first-order differences, with $\mathbf{v}_1 = \mathbf{0}$, capturing the rate structure of the gait cycle [30].

TABLE IV
COMPOSITION OF THE 78-DIMENSIONAL PER-FRAME DESCRIPTOR

Block	Dims	Index	Content
Norm. coordinates	36 (12 × 3)	[0:36)	Hip-centred (x, y, z)
Joint angles	6	[36:42)	Knee/hip/ankle flexion
Velocities	36 (12 × 3)	[42:78)	Δ coordinates
Total	**78**		

3) Temporal and Statistical Normalisation: Sequences are resampled to $T = 60$ frames: longer sequences are uniformly subsampled, shorter ones zero-padded, and clips yielding fewer than 10 valid frames are rejected. At 30 fps, 60 frames span roughly two seconds, covering one to two complete gait cycles [13]. Each of the 78 dimensions is then z-score normalised, $\hat{f}_j = (f_j - \mu_j)/\max(\sigma_j, 10^{-6})$, which is necessary because the blocks occupy incommensurate scales [31]. Under LOOCV, μ_j and σ_j are recomputed inside every fold.

We record one protocol detail for reproducibility. The evaluation script that produced our headline LOOCV figures normalises the held-out subject using that subject's own statistics rather than the training split's—a transductive choice, and not the one a strict reading of the protocol would imply. We measured what it is worth by running the deployed model both ways across all 13 folds: training statistics yield $94.67\% \pm 3.06\%$ ROC-AUC against $94.97\% \pm 2.90\%$ for the shipped behaviour, a difference of 0.30 points that is not statistically significant ($p = 0.73$, paired, $n = 13$). The headline result is therefore insensitive to this choice, and all ablation comparisons in Section V-D use training statistics throughout.

D. Identity Enrolment

Each identity I is enrolled by averaging the descriptors of its original (non-augmented) recordings per timestep,

$$\bar{\mathbf{f}}_t^{(I)} = \frac{1}{|V_I|} \sum_{v \in V_I} \mathbf{f}_t^{(v)}, \qquad t = 1, \ldots, 60, \tag{5}$$

yielding a 60×78 signature stored per subject. Averaging across recordings suppresses per-take noise while preserving the subject's characteristic temporal profile. The signature is used as a full sequence, not collapsed to a vector—the verification network consumes it as a second input stream.

E. Verification Network

The verification network classifies the discrepancy between the observed sequence $\mathbf{V} \in \mathbb{R}^{T \times 78}$ and the claimed identity's signature $\mathbf{C} \in \mathbb{R}^{T \times 78}$ directly, without first reducing either to a pooled embedding.

1) Pairwise Comparison Tensor: Three per-timestep comparison signals are concatenated,

$$\mathbf{F} = [\,(\mathbf{V} - \mathbf{C}) \,\|\, |\mathbf{V} - \mathbf{C}| \,\|\, (\mathbf{V} \odot \mathbf{C})\,] \in \mathbb{R}^{T \times 234}, \tag{6}$$

where the signed difference localises *where* and *in which direction* the sequences diverge, the absolute difference supplies magnitude invariant to sign, and the elementwise product responds where the two agree.

Algorithm 1 Difference-Based Verification Forward Pass

Require: Observed $\mathbf{V} \in \mathbb{R}^{B \times T \times 78}$, claimed $\mathbf{C} \in \mathbb{R}^{B \times T \times 78}$
1: $\mathbf{D} \leftarrow \mathbf{V} - \mathbf{C}$
2: $\mathbf{F} \leftarrow [\mathbf{D} \,\|\, |\mathbf{D}| \,\|\, \mathbf{V} \odot \mathbf{C}] \in \mathbb{R}^{B \times T \times 234}$
3: $\mathbf{F} \leftarrow \mathrm{permute}(\mathbf{F}) \in \mathbb{R}^{B \times 234 \times T}$
4: $\mathbf{F} \leftarrow \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv1D}_{234 \to 64,\, k=7}(\mathbf{F})))$; dropout
5: $\mathbf{F} \leftarrow \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv1D}_{64 \to 64,\, k=5}(\mathbf{F})))$; dropout
6: $\mathbf{F} \leftarrow \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv1D}_{64 \to 32,\, k=3}(\mathbf{F})))$
7: $\mathbf{z} \leftarrow \mathrm{AdaptiveAvgPool1D}(\mathbf{F}) \in \mathbb{R}^{B \times 32}$
8: $\boldsymbol{\ell} \leftarrow \mathrm{Linear}_{32 \to 2}(\mathrm{Dropout}(\mathrm{ReLU}(\mathrm{Linear}_{32 \to 32}(\mathbf{z}))))$
9: **return** $P(\text{AUTHENTIC}) = \mathrm{softmax}(\boldsymbol{\ell})_1$

2) Temporal Classifier: $\mathbf{F}$ is transposed to channel-first order and passed through three 1D convolutional stages with decreasing receptive fields ($234 \to 64$, $k = 7$; $64 \to 64$, $k = 5$; $64 \to 32$, $k = 3$), each with batch normalisation and ReLU, and dropout on the first two. Global average pooling over time yields a 32-dimensional summary, which a two-layer MLP maps to two logits. The verdict score is $P(\text{AUTHENTIC}) = \mathrm{softmax}(\cdot)_1$. Algorithm 1 states the forward pass.

3) Why Differences Rather Than Embedding Distances: Our initial design followed the standard Siamese template: encode both sequences with shared weights, pool to embeddings, and score their distance. On this corpus it collapsed—the encoder converged to a near-constant embedding, driving all pairwise distances towards a single value and the classifier towards the majority prediction. With 13 identities and a per-timestep signature that is itself an average of few recordings, the pooling step discards precisely the fine temporal structure that separates individuals. Comparing sequences before pooling avoids the failure: the discriminative information is preserved at full temporal resolution and the pooling that does occur happens after the comparison, on evidence rather than on identity.

F. Auxiliary Representation Branch

The trained checkpoint also contains a representation branch, retained from the Siamese design and used for enrolment analysis and diagnostics: a residual 1D CNN encoder (GAITENCODER, $78 \to 64 \to 128$ with two residual blocks [32]) feeding a dual-path temporal model (DUALPATHTEMPORALMODEL) whose BiLSTM [33] ($h = 64$, one layer, bidirectional) and Transformer encoder [34] ($d_{\text{model}} = 128$, 4 heads, 2 layers, $d_{\text{ff}} = 512$, GELU [35], pre-LayerNorm [36], sinusoidal positional encoding) are fused by an MLP into a 128-dimensional embedding. Linear layers use Xavier initialisation [37] and convolutions He initialisation [38]. Table V gives the configuration as trained.

We must be explicit about this branch's role, because the distinction affects how the results in Section V should be read. Under the cross-entropy objective applied to the verification logits, gradients reach only the pairwise-comparison path. The representation branch produces an embedding that is not consumed by the classifier and receives no learning signal from the training objective. Of the model's $848{,}614$ parameters, $133{,}058$ (15.7%) lie on the decision path.

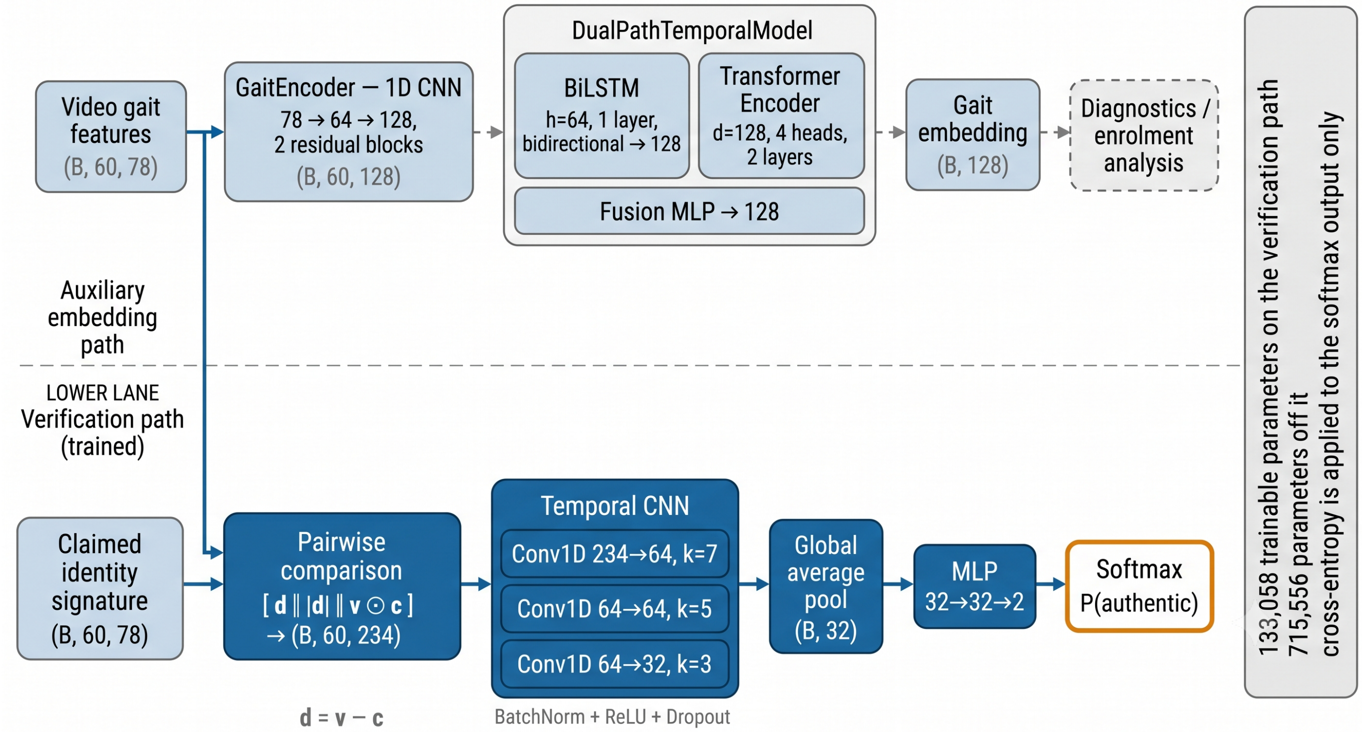

Fig. 4. Network architecture as trained. The lower lane is the verification path: the observed and claimed sequences are combined into a 234-channel per-timestep comparison tensor, passed through a three-layer temporal CNN, pooled and classified. The upper lane—a residual 1D CNN encoder feeding a parallel BiLSTM and Transformer—produces a 128-dimensional gait embedding used for enrolment analysis and diagnostics. The two lanes do not join: under the cross-entropy objective applied to the softmax output, 133,058 parameters lie on the verification path and 715,556 do not. Section III-F states the consequences of this separation, and Section VII-C treats it as a threat to validity.

TABLE V
MODEL CONFIGURATION AS TRAINED

Component	Setting	Params
Verification path (trained)		
Comparison tensor	$78 \times 3 = 234$ channels	—
Temporal CNN	$k = 7, 5, 3$; 64, 64, 32 ch.	131,936
Classifier MLP	$32 \to 32 \to 2$	1,122
Auxiliary branch (not on decision path)		
GaitEncoder	$78 \to 64 \to 128$	129,600
DualPathTemporalModel	BiLSTM h=64; Trans. d=128	546,304
Identity verifier head	128, hidden 64	31,266
Standalone classifier	$128 \to 64 \to 2$	8,386
Total		**848,614**
On decision path		**133,058**

TABLE VI
TRAINING CONFIGURATION

Parameter	Value	Rationale
Optimiser	AdamW	Decoupled decay [40]
Learning rate	1×10^{-3}	Adam default [44]
Weight decay	1×10^{-4}	Moderate regularisation
Scheduler	Plateau, $\times 0.5$	Patience 7
Grad. clipping	$\|g\| \leq 1.0$	Stability [41]
Dropout	0.1	Small corpus [42]
Batch size	16	$\sim$1k samples [45]
Max epochs	50 (LOOCV: 30)	With early stopping
Early stopping	Patience 20	[43]
Loss	Weighted CE	Two-class logits
Pair sampling	50/50 balanced	[39]
Hardware	NVIDIA RTX 3050, 6 GB	CUDA 12.4

All quantitative results in this paper are therefore properties of the difference-based verification network, not of the CNN–BiLSTM–Transformer stack. Section V-D takes the further step of asking whether that stack *should* be connected, and finds that it should not: every configuration that routes it onto the decision path is significantly worse than leaving it off. Fig. 4 draws the separation explicitly.

G. Training

Training pairs are sampled with a 50/50 balance: half pair a video with its own identity's signature (label AUTHENTIC), half with a uniformly sampled different identity's signature (label DEEPFAKE). This simulates the identity-mismatch condition a face swap induces, and prevents the degenerate constant-prediction solution [39]. Optimisation uses AdamW [40] at 10^{-3} with weight decay 10^{-4}, batch size 16, gradient-norm clipping at 1.0 [41], dropout 0.1 [42], and `ReduceLROnPlateau` (factor 0.5, patience 7). Early stopping monitors validation accuracy with patience 20 and $\delta_{\min} = 10^{-3}$ [43]. The loss is class-weighted cross-entropy on the two verification logits. Table VI summarises the settings; the single-split reference model reached 93.92% validation accuracy at epoch 46, against 98.38% training accuracy.

Algorithm 2 Leave-One-Subject-Out Evaluation

Require: Corpus $\mathcal{D}$ over subjects $\{s_1, \ldots, s_{13}\}$

$\mathcal{R} \leftarrow \varnothing$
for $k = 1$ **to** 13 **do**
  $\mathcal{D}_{\text{te}} \leftarrow \{(\mathbf{x}, y) : \text{subj}(\mathbf{x}) = s_k\}$
  $\mathcal{D}_{\text{tr}} \leftarrow \mathcal{D} \setminus \mathcal{D}_{\text{te}}$
  $(\mu, \sigma) \leftarrow \text{stats}(\mathcal{D}_{\text{tr}})$ {training split only}
  Re-enrol the 12 training identities from $\mathcal{D}_{\text{tr}}$
  Train a fresh model on $\mathcal{D}_{\text{tr}}$ for 30 epochs
  Score $\mathcal{D}_{\text{te}}$ with balanced genuine/impostor pairs
  $\mathcal{R} \leftarrow \mathcal{R} \cup \{\text{Acc}, \text{F1}, \text{P}, \text{R}, \text{AUC}, \text{EER}\}_k$
end for
return per-fold $\mu \pm \sigma$, and pooled metrics over all $\sum_k n_k = 2{,}240$ scores

IV. Experimental Setup

A. Leave-One-Subject-Out Cross-Validation

The primary protocol is 13-fold leave-one-subject-out cross-validation (Algorithm 2). In fold k, every recording of subject s_k—original and all 15 augmentations—is withheld, normalisation statistics are recomputed from the remaining 12 subjects, and a model is trained from scratch for 30 epochs before being evaluated on the held-out subject's pairs.

Subject-level rather than video-level partitioning is not a refinement but a requirement. Because augmented clips are derived from originals, a video-level split would place 15 near-duplicates of a test clip into training and produce an estimate with no bearing on unseen subjects. LOOCV is the appropriate protocol for small-cohort biometric evaluation [17], with the caveat that its variance is high at this cohort size [46]. The full 13-fold procedure completed in 1,228 s.

B. Metrics and Operating Points

We report ROC-AUC as a threshold-free measure of separability [47]; equal error rate, the biometric convention, at the point where false accept and false reject rates coincide [17]; and accuracy, precision, recall and F1 at an explicit operating point.

Two aggregation modes are reported throughout and should not be conflated. The *per-fold* statistic is the mean and standard deviation of a metric computed independently within each of the 13 folds; it describes how much performance varies by subject. The *pooled* statistic is computed once over all 2,240 scores concatenated across folds; it describes the system's behaviour on the population. The two differ—pooled AUC is 94.95% against a per-fold mean of 95.10%—because folds carry unequal sample counts and because pooling forces a single global ranking across models trained on different splits.

Two operating points are likewise reported. The native $\tau = 0.5$ argmax decision is the point at which the per-fold accuracy, precision, recall and F1 in the results tables were computed. The deployed threshold is instead chosen by Youden's J [48], [49],

$$\tau^* = \arg\max_{\tau} \left[\text{TPR}(\tau) - \text{FPR}(\tau)\right], \tag{7}$$

evaluated on the pooled LOOCV ROC, giving $\tau^* = 0.7737$ with TPR $= 83.57\%$, FPR $= 8.48\%$ and $J = 0.7509$. The threshold is derived from data, not assigned.

C. Ablation Design

Five variants were trained under the full leave-one-subject-out protocol at three random seeds, differing only in how the observed and claimed sequences are encoded before comparison. All share folds, seeds, schedule, and the downstream temporal CNN and classifier, so each can be compared to the deployed model pairwise over $13 \times 3 = 39$ matched observations. The control is an identity encoder, which reproduces the deployed model exactly; Section V-D reports the design and the outcome.

D. Face-Swap Test Protocol

Face-swapped clips were generated with FaceFusion [7] using the InSwapper `inswapper_128_fp16` model [8] under a face-only mask that preserves body, neck and shoulders. Each clip composites one enrolled subject's face onto another enrolled subject's walking footage and is named `{BodyPerson}_body_{FacePerson}_face.mp4`. Presented with the face subject as claimed identity, a correct system should reject, because the locomotion belongs to the body subject.

The corpus contains three such clips. Gait preservation through the swap is assessed by comparing MediaPipe features before and after manipulation under three criteria—PCK@0.05 $\geq$ 90% [50], mean cosine similarity of hip-normalised pose vectors ≥ 0.95 [16], and Pearson correlation of step-width and symmetry time series ≥ 0.85 [51]—with a majority vote across the three [52].

V. Results

A. Aggregate Performance

Table VII reports the 13-fold aggregate. The system separates genuine from impostor pairs with a pooled ROC-AUC of 94.95%, and at the native operating point classifies $87.04\% \pm 3.65\%$ of held-out pairs correctly with an F1 of $87.12\% \pm 3.77\%$. The equal error rate is $12.27\% \pm 3.80\%$ per fold and 12.77% pooled.

Recall carries visibly the largest dispersion (± 6.82), roughly twice that of AUC. Since these are subject-held-out folds, this says that how readily a given individual's genuine pairs are accepted depends more on who that individual is than overall separability does—a point Section V-B develops.

Fig. 5 shows the pooled ROC alongside all 13 fold curves. The per-fold envelope is tight over most of the operating range and widens in the low-FPR region, which is where a deployed system would be tuned—a caution worth carrying into Section VII.

B. Per-Fold Dispersion

Table VIII and Fig. 6 give the fold-level detail that an aggregate $\pm\sigma$ suppresses. The distribution is unimodal with a single clear low outlier: Subject 12 yields 87.00% AUC and

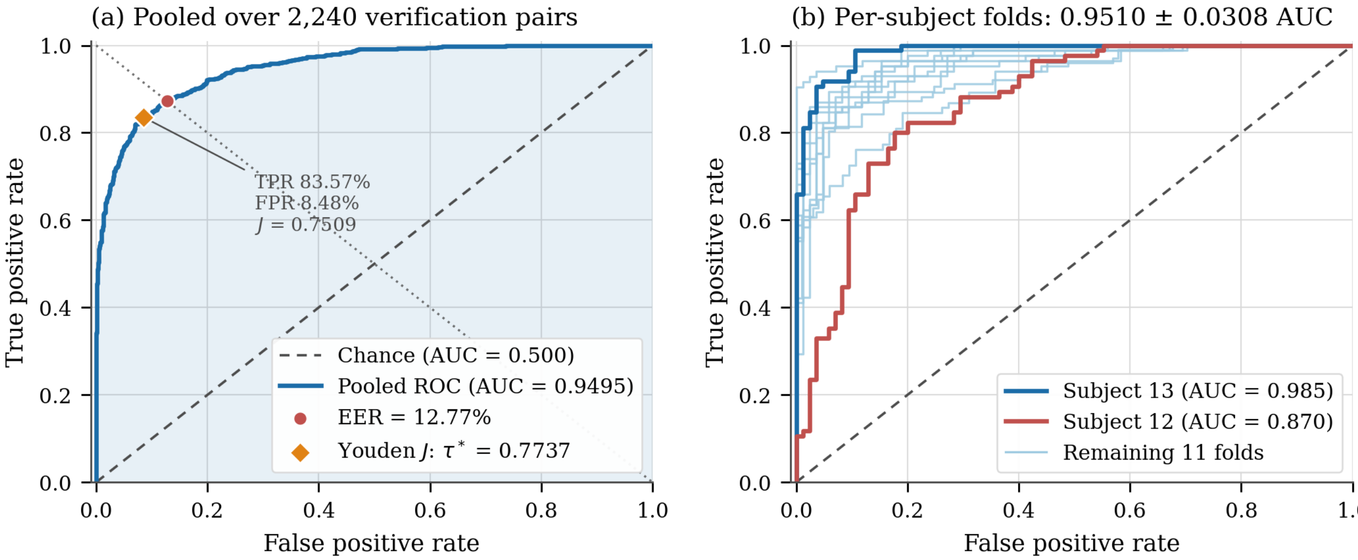

Fig. 5. ROC analysis of the leave-one-subject-out evaluation. (a) Pooled over all 2,240 verification pairs, with the equal-error point and the Youden-optimal operating point $\tau^* = 0.7737$ marked; the two sit far apart, which is why the operating point must be reported alongside the AUC rather than assumed. (b) All 13 subject-held-out folds overlaid, with the best (Subject 13, 0.985) and worst (Subject 12, 0.870) drawn in full. The envelope is narrow through the mid-range and widens near the origin, indicating that low-false-accept operation is the regime where subject identity matters most.

TABLE VII
LOOCV AGGREGATE PERFORMANCE OVER 13 SUBJECT-HELD-OUT FOLDS AND 2,240 VERIFICATION PAIRS

Metric	**Per-fold** $\mu \pm \sigma$	**Pooled**
ROC-AUC (%)	95.10 ± 3.08	**94.95**
Accuracy (%)	87.04 ± 3.65	87.01
F1 (%)	87.12 ± 3.77	87.15
Precision (%)	86.51 ± 4.72	86.20
Recall (%)	88.23 ± 6.82	88.12
EER (%)	12.27 ± 3.80	12.77
Youden-optimal operating point (pooled ROC)		
Threshold τ^*	0.7737	
TPR / FPR at τ^*	83.57% / 8.48%	
Youden J	0.7509	

TABLE VIII
PER-FOLD LOOCV RESULTS (HELD-OUT SUBJECT; %)

Subj.	**Acc**	**F1**	**Prec**	**Rec**	**AUC**	**EER**	n
S1	89.29	90.00	84.38	96.43	98.16	6.55	168
S2	87.65	87.72	87.21	88.24	95.04	12.94	170
S3	88.24	88.10	89.16	87.06	95.54	11.76	170
S4	82.14	81.01	86.49	76.19	90.87	19.05	168
S5	84.80	84.26	87.37	81.37	92.59	15.69	204
S6	86.47	84.97	95.59	76.47	96.71	9.41	170
S7	88.24	88.76	84.95	92.94	95.70	9.41	170
S8	88.24	88.64	85.71	91.76	96.64	12.94	170
S9	86.47	87.57	81.00	95.29	95.71	11.76	170
S10	90.00	89.94	90.48	89.41	97.20	9.41	170
S11	90.00	90.50	86.17	95.29	96.62	11.76	170
S12	77.65	78.65	75.27	82.35	87.00	20.00	170
S13	92.35	92.49	90.91	94.12	98.53	8.82	170
Mean	87.04	87.12	86.51	88.23	95.10	12.27	
Std	3.65	3.77	4.72	6.82	3.08	3.80	

77.65% accuracy, roughly 2.6 standard deviations below the mean accuracy, while the remaining twelve folds fall between 82.14% and 92.35%. Removing that fold would raise mean accuracy to 87.83%; we report it included.

Two structural observations follow. First, the ordering of folds is stable across metrics—the subjects that are hard by AUC are hard by accuracy, F1 and EER alike—which points to a property of those subjects' gait rather than to threshold artefacts. Second, precision and recall trade off across folds in opposite directions (Subject 6: 95.59% precision, 76.47% recall; Subject 9: 81.00% precision, 95.29% recall), meaning a single global threshold sits at a different point on each subject's curve. Per-subject calibration is a natural extension we did not pursue.

C. Operating Points and Error Structure

Fig. 7 contrasts the two operating points, and the difference between them matters for deployment. At the native $\tau = 0.5$ the system produces 987 true positives, 962 true negatives, 158 false positives and 133 false negatives—balanced errors, 87.01% accuracy. Moving to the Youden-optimal $\tau^* = 0.7737$ shifts 63 pairs from false-positive to true-negative at the cost of 51 additional false negatives: false positives fall from 158 to 95 and precision rises from 86.20% to 90.79%, while recall falls from 88.12% to 83.57%.

Which of these is preferable is a policy question, not a modelling one. In a forensic screening context a false positive means accepting a forgery, and the stricter threshold is the right default. We flag one inconsistency in earlier descriptions of this system for the record: the frequently quoted 987/962/158/133 confusion matrix is the $\tau = 0.5$ matrix, not the Youden-threshold matrix, though it has sometimes been reported alongside $\tau^* = 0.7737$.

The score distribution in Fig. 8 explains why the two thresh-

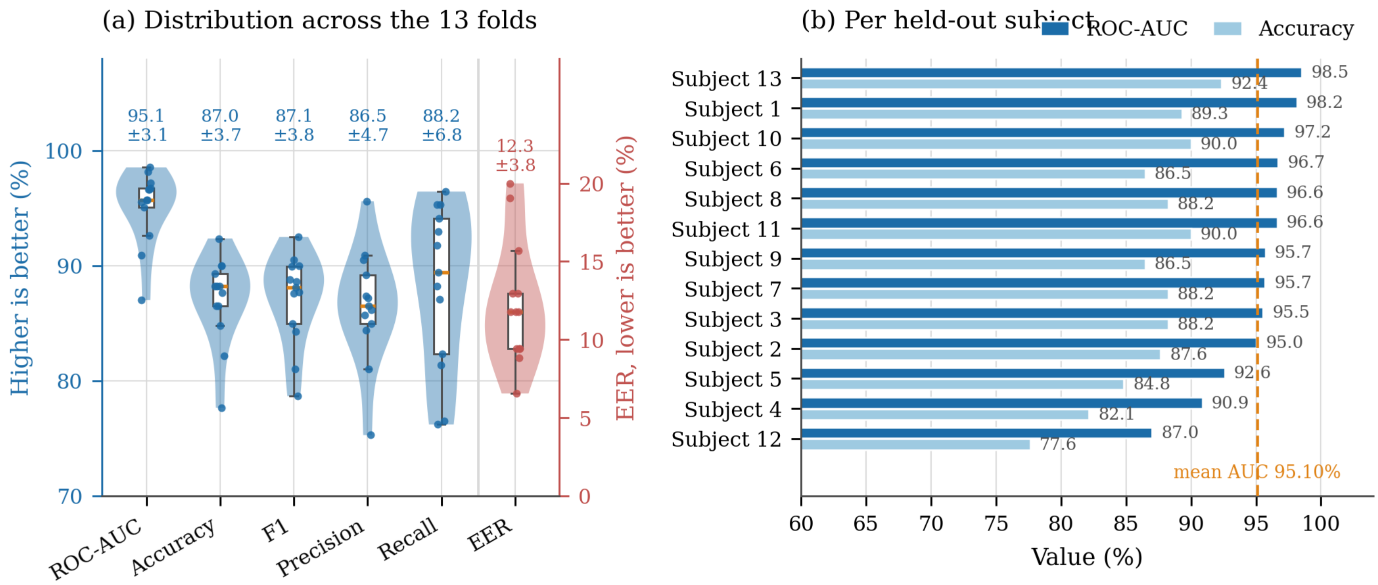

Fig. 6. Fold-to-fold dispersion across the 13 subject-held-out folds. (a) Distribution of each metric, with individual folds overplotted; EER is drawn against its own right-hand scale because its 6.6–20.0% range would otherwise compress the remaining five into an unreadable band. The distributions are unimodal, confirming that the reported $\pm\sigma$ summarises a coherent spread rather than concealing a split population. (b) Per-subject ROC-AUC and accuracy, ordered by AUC. One fold (Subject 12) sits clearly below the rest on every metric; the remaining twelve span roughly ten accuracy points.

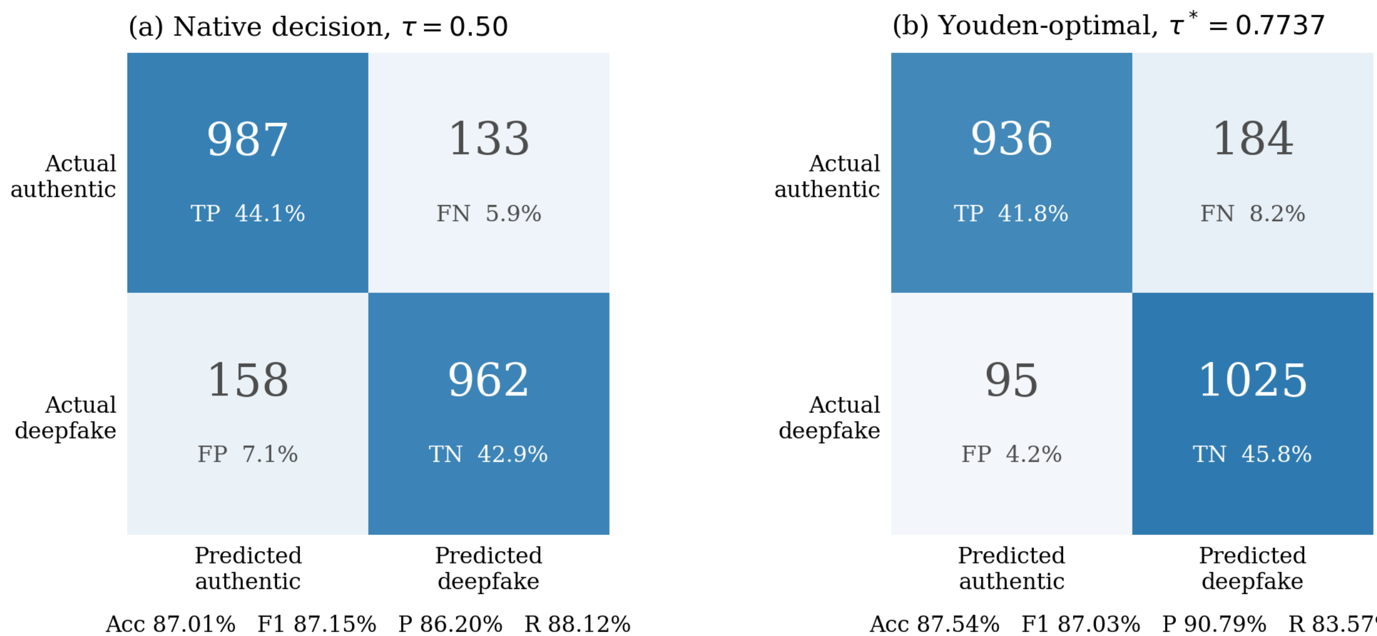

Fig. 7. Pooled confusion matrices over all 2,240 LOOCV verification pairs at the two operating points. (a) The network's native $\tau = 0.5$ argmax decision, which is the point at which the per-fold metrics of Table VIII were computed. (b) The Youden-optimal $\tau^* = 0.7737$ actually applied at inference. Tightening the threshold converts 63 false accepts into correct rejections at the cost of 51 additional false rejections, raising precision from 86.20% to 90.79% while lowering recall from 88.12% to 83.57%. The two matrices should not be quoted interchangeably.

olds differ so much while yielding similar accuracy. Both classes are strongly saturated—genuine pairs have median score 0.999, impostor pairs 0.0007—so the region between the two thresholds contains only 86 pairs, 3.8% of the total. The classifier is confident almost everywhere and the threshold choice redistributes a small, hard residue.

D. Architectural Ablation

The dual-path stack of Section III-F is inert in the deployed checkpoint. The obvious question is whether it should be: would wiring it onto the decision path improve verification? Table IX and Fig. 9 answer it.

We construct five variants that differ only in how the observed and claimed sequences are encoded before they are compared. Each encodes both sequences with shared

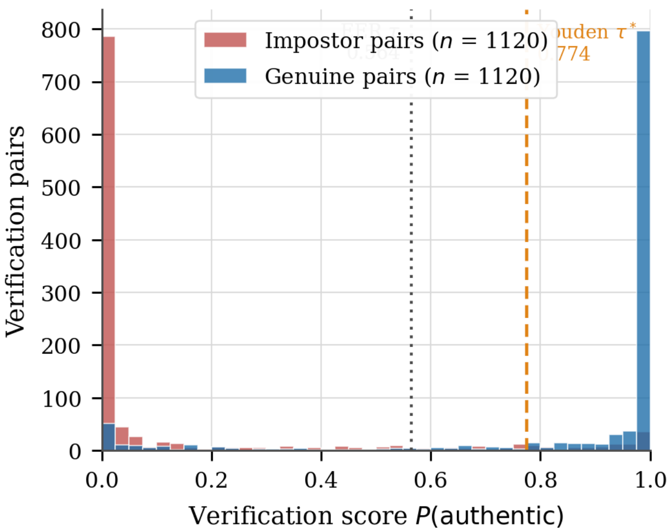

Fig. 8. Distribution of $P(\text{AUTHENTIC})$ over the 2,240 pooled verification pairs. Both classes saturate against their extremes (genuine median 0.999, impostor median 0.0007), leaving only 86 pairs—3.8% of the total—between the equal-error and Youden thresholds. The model is decisive on the large majority of pairs, and threshold selection governs the disposition of a small ambiguous residue rather than shifting bulk behaviour. The corollary is that reported confidence values should not be read as calibrated probabilities.

weights, preserves the time axis, and forms the same per-timestep comparison tensor, which the same temporal CNN and classifier then consume. The control, *Raw*, uses an identity encoder and therefore reproduces the deployed model exactly at 133,058 parameters. Four *Raw + X* variants concatenate the encoded comparison onto the raw one, so each contains the deployed model's evidence and can only justify itself by improving on it. *Hybrid only* removes the raw comparison, isolating what the encoder contributes alone. Every variant is trained under the full leave-one-subject-out protocol at three random seeds, giving $13 \times 3 = 39$ observations per variant on identical folds and identical initialisations.

The result is unambiguous and it is negative. The deployed model is the best of the five, at $94.25\% \pm 3.09\%$ ROC-AUC. Every encoder variant is worse, and because all variants share folds and seeds the comparison can be made pairwise, which removes fold difficulty and initialisation luck. The paired deficits range from -2.53 points for *Raw + CNN* to -5.93 for *Raw + BiLSTM*; all four are significant at $p < 10^{-3}$ (two-sided paired t-test, $n = 39$), and no variant wins more than 9 of its 39 paired trials. The 95% confidence interval on each deficit lies wholly below zero (Fig. 9b). Adding capacity does not merely fail to help: the largest variant, at $981{,}000$ parameters, is 4.22 points worse than the $133{,}058$-parameter control.

Hybrid only settles which half of the architecture carries the signal. Stripped of the raw per-timestep comparison, it reaches $50.70\% \pm 20.81\%$ ROC-AUC—indistinguishable from chance, and with a standard deviation four times that of any other variant, indicating it often fails to learn at all. The discriminative information is in the direct per-dimension comparison between the observed sequence and the enrolled signature, where dimension j of each is the same physical quantity and subtraction is meaningful. Passing both through

TABLE IX
LOOCV Ablation Over 13 Folds and 3 Seeds ($n = 39$ Paired Observations per Variant). Δ is the Paired Change in ROC-AUC Against the Deployed Model

Variant	Params	ROC-AUC (%)	Acc (%)	Δ AUC
Raw (deployed)	133,058	$\mathbf{94.25 \pm 3.09}$	**86.04**	—
Raw + CNN	434,690	91.72 ± 4.13	82.66	-2.53^*
Raw + Transformer	712,002	90.37 ± 5.13	80.32	-3.88^*
Raw + Hybrid	980,994	90.03 ± 6.11	81.26	-4.22^*
Raw + BiLSTM	379,074	88.32 ± 6.08	74.84	-5.93^*
Hybrid only	876,162	50.70 ± 20.81	49.37	-43.55^*

*Significant at $p < 10^{-3}$, two-sided paired t-test over the 39 matched (fold, seed) pairs. "Hybrid only" omits the raw comparison entirely.

a learned encoder first destroys that correspondence and forces the classifier to recover it from a representation that begins as a random projection.

Two consequences follow. First, the ablation retroactively justifies the deployed architecture: the inert branch of Section III-F is not a missed opportunity, and connecting it would cost accuracy. Second, the result is a caution against the assumption that a richer temporal encoder is a free improvement. On this task, with this cohort size, it is a reliable regression.

E. Face-Swap Video Validation

Three face-swapped clips were generated as described in Section IV, pairing enrolled subjects so that both the body source and the face source have enrolled signatures. The mechanism the protocol tests is not in question: an InSwapper-class pipeline composites through a mask confined to the facial region, leaving the skeletal trajectory of the body subject intact by construction, so the gait recovered from a forged clip is the body subject's. Each clip was scored against the claimed (face) identity using checkpoint_epoch_46_best.pth at the LOOCV Youden threshold $\tau^* = 0.7737$, and independently re-scored against every enrolled identity to identify the closest match.

All three clips were correctly rejected. Table X reports the similarity to the claimed identity, the identity the system actually matched, and the similarity to that match. In every case the claimed identity scores below 0.006—four to six orders of magnitude below τ^*—while the true body source scores above 0.9998. The system does not merely fail to confirm the claimed identity; it confidently identifies the correct one, exactly as the threat model of Section III-A predicts. Frame extraction succeeded on the full length of all three clips (135, 149 and 327 valid frames respectively), so the result is not an artefact of partial pose failure on manipulated footage.

One result merits a caveat rather than a footnote. For Subject 8_body_Subject 5_face, the correct match (Subject 8, 0.99999) is nearly tied by an impostor score for Subject 10 (0.99986)—the two nearest scores in the entire validation, four orders of magnitude closer than any other pair in Table X. The system still ranks the true body source first and still rejects the claimed identity by a wide margin, so the clip is correctly handled; but the near-collision shows that the embedding space is not uniformly well separated across all subject pairs, and a larger validation set would be needed to establish how often this occurs.

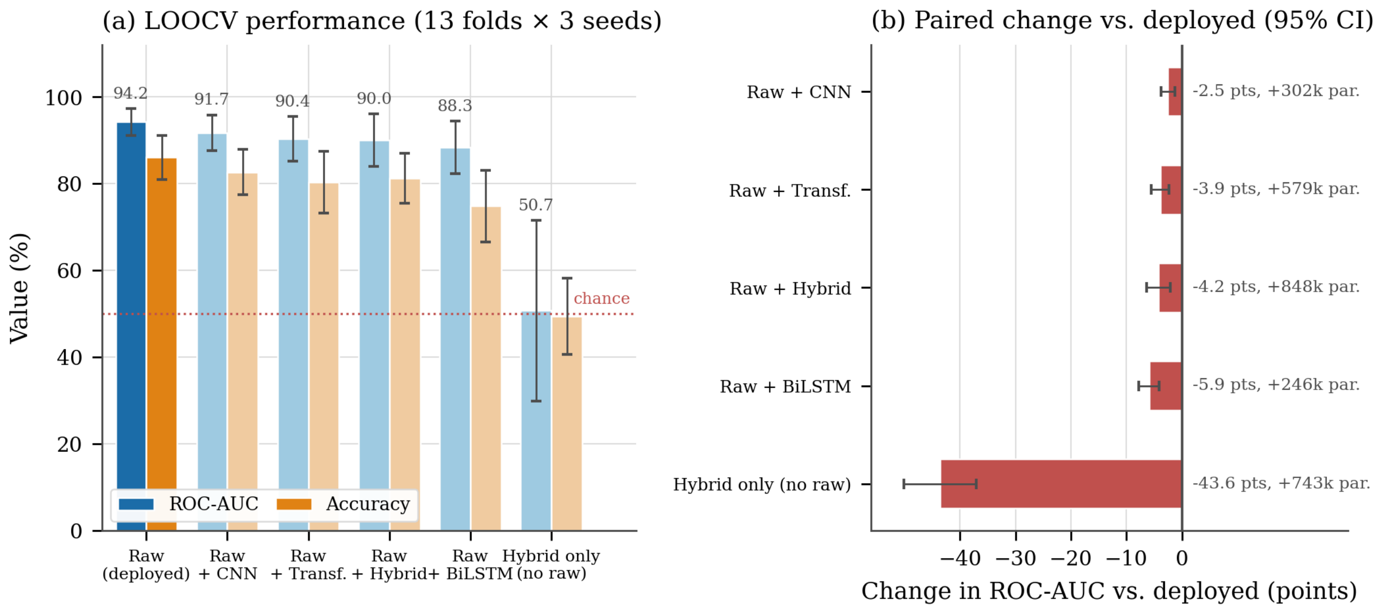

Fig. 9. LOOCV ablation over five variants, each measured on 13 subject-held-out folds at 3 seeds. (a) ROC-AUC and accuracy per variant, with standard deviation across the 39 observations; the deployed model (dark) is best, and removing the raw per-timestep comparison entirely (rightmost) lands at chance. (b) Because every variant shares folds and seeds with the control, the comparison is made pairwise, removing fold difficulty and initialisation luck. All five 95% confidence intervals lie wholly below zero. The deficit does not shrink as parameters are added: the largest variant is 4.22 points worse than a control one-seventh its size.

TABLE X
FACE-SWAP VIDEO VALIDATION ($n = 3$; EXISTENCE CHECK, NOT A STATISTICALLY POWERED RESULT)

Clip (body_face)	Claimed	Sim.	Matched	Sim.
Subj.6_Subj.3	Subj. 3	0.0057	Subj. 6	0.9999
Subj.7_Subj.11	Subj. 11	0.0003	Subj. 7	0.9999
Subj.8_Subj.5	Subj. 5	<0.0001	Subj. 8	0.99999*

*Nearest impostor score for this clip (Subject 10) was 0.99986, the closest margin observed; see discussion in text.

We report the scope of this result precisely rather than overstate it. Three clips, drawn from a single face-swap pipeline (FaceFusion / InSwapper) under one masking configuration, is an existence check confirming the mechanism behaves as the threat model predicts—it is not a statistically powered validation, and it says nothing about behaviour under other face-swap models, face-enhancement post-processing, or heavier recompression. Section VII-C states what a properly powered version of this experiment would require.

VI. EXPLAINABILITY

A. Attribution Method

To determine what the verification network uses, we compute gradient-times-input attribution on the 78-dimensional descriptor. For a sample scored towards class c, we backpropagate the corresponding logit to the input and take

$$A_{t,j} = \left| \hat{f}_{t,j} \cdot \frac{\partial \ell_c}{\partial \hat{f}_{t,j}} \right|, \tag{8}$$

aggregating over timesteps and over the three coordinates belonging to each joint, then normalising to $[0, 1]$. We additionally compute Grad-CAM [53] over the final convolutional layer for temporal localisation. Attribution is averaged over 26 samples, two per identity.

Because the gradient path runs through the verification head, this attribution describes the function that actually produces the verdict. We use the term gradient-times-input rather than Grad-CAM for the joint analysis, since the former is what the joint-level numbers are computed with [54], [55].

B. Joint-Level Attribution

Fig. 10 and Table XI give the ranking. The left shoulder receives the highest attribution (1.000), followed by the right heel (0.940), left foot index (0.931) and left knee (0.897); both hips rank lowest (left 0.665, right 0.566).

The structure is biomechanically coherent. The two extremes of the kinematic chain dominate: shoulders, which carry the trunk countermotion that balances angular momentum during swing, and the distal foot segments, which register heel strike and toe-off—the discrete, individually-variable events that define the gait cycle [13], [14]. The hips rank lowest, which is expected rather than anomalous: the pelvis centre is the origin of the coordinate normalisation, so hip positions are near-constant by construction and carry little residual variance.

A left–right asymmetry is visible (left shoulder 1.000 against right 0.873; left foot 0.931 against right 0.766; but right heel 0.940 against left 0.709). We do not over-interpret it. With 13 subjects and 26 attribution samples it is as consistent with cohort-specific gait asymmetry, or with viewpoint bias between the frontal and side recordings, as with anything the model has learned about laterality in general.

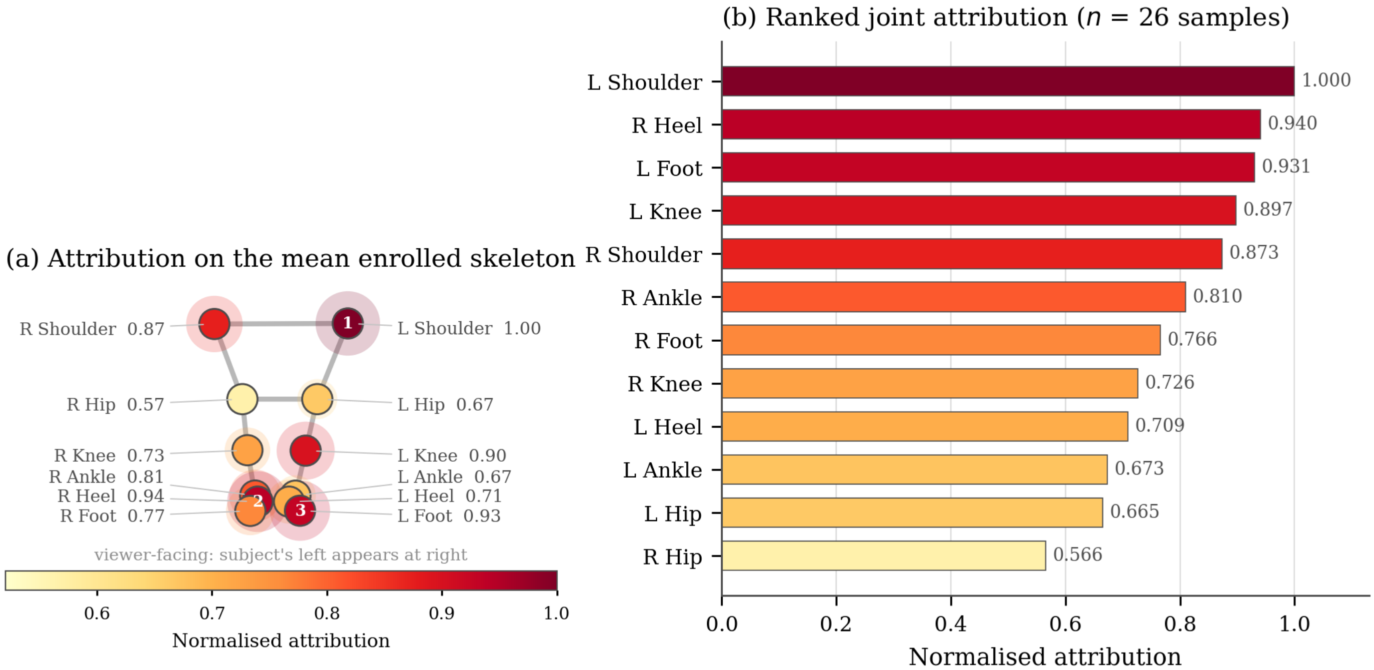

Fig. 10. Gradient-times-input attribution over the 12 gait keypoints, averaged across 26 samples (two per identity). (a) Attribution rendered on a skeleton whose geometry is the mean of all 13 enrolment signatures across their 60 timesteps, so the layout is data-derived; marker size and colour both encode attribution. (b) The same values ranked. Attribution concentrates at the two ends of the kinematic chain—shoulder countermotion and distal heel-strike and toe-off structure—while the hips rank lowest, as expected given that the pelvis centre is the origin of the coordinate normalisation and therefore carries little residual variance.

TABLE XI
JOINT ATTRIBUTION RANKING (NORMALISED, 26 SAMPLES)

Rank	Joint	Score	Rank	Joint	Score
1	Left shoulder	1.000	7	Right foot index	0.766
2	Right heel	0.940	8	Right knee	0.726
3	Left foot index	0.931	9	Left heel	0.709
4	Left knee	0.897	10	Left ankle	0.673
5	Right shoulder	0.873	11	Left hip	0.665
6	Right ankle	0.810	12	Right hip	0.566

TABLE XII
ATTRIBUTION BY FEATURE GROUP, AGAINST DIMENSIONAL SHARE

Group	Attrib. (%)	Dims	Dim. share (%)	Ratio
Normalised coordinates	47.68	36	46.15	1.03
Velocities	37.40	36	46.15	0.81
Joint angles	14.92	6	7.69	**1.94**

C. Feature-Group Attribution

Table XII and Fig. 11 decompose attribution by feature block. Coordinates take 47.68%, velocities 37.40% and joint angles 14.92%. Read as raw shares this simply says that coordinates matter most, which is unsurprising given that they occupy 36 of 78 dimensions.

The informative comparison is against dimensional share. Coordinates and velocities each occupy 46.15% of the input and draw 47.68% and 37.40% of attribution respectively—ratios of 1.03 and 0.81. Joint angles occupy 7.69% of the input and draw 14.92%: a ratio of 1.94. Per dimension, the six angle features are roughly twice as influential as the average coordinate and nearly two and a half times as influential as the average velocity.

This is the clearest practical finding in the attribution analysis. The six scale-free, rotation-invariant biomechanical constraints are the most information-dense part of the descriptor, which suggests that a cheaper feature set weighted towards angular kinematics may retain most of the discriminative power—directly relevant to the embedded deployment discussed in Section VII. Within the angle block (Fig. 11b), the left ankle (1.000) and left knee (0.896) dominate, consistent with the distal emphasis in the joint-level ranking.

VII. DISCUSSION

A. Comparative Positioning

Table XIII situates this work against the literature of Section II. We stress that the table is not a ranking. Every row is measured on a different corpus under a different protocol against a different task, and our own figure is a leave-one-subject-out result on a 13-subject corpus collected by the authors—a fundamentally different quantity from a cross-dataset AUC on Celeb-DF. Reading our 94.95% as exceeding AltFreezing's 89.5% would be a category error, and we do not make that claim. The table's purpose is to show where the modality and task sit relative to established work.

Gait evidence has also begun to appear in forensic practice: a commercial gait and body-structure analysis was reported in 2025 to have been accepted as evidence in a European

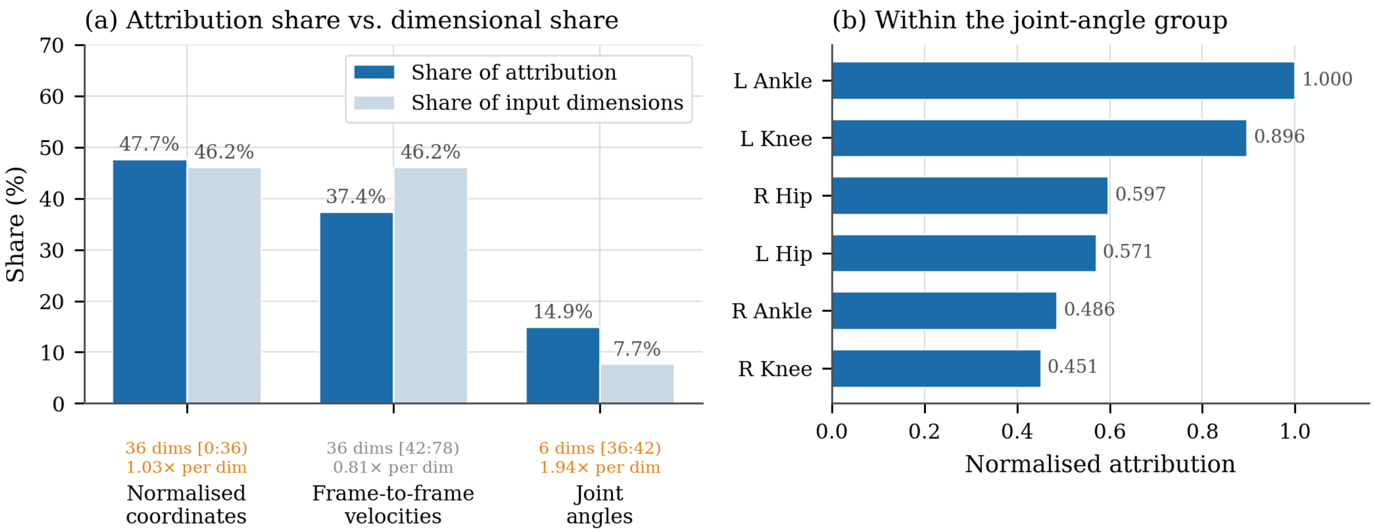

Fig. 11. Attribution by feature group. (a) Each group's share of total attribution against its share of the 78 input dimensions. Coordinates and velocities draw roughly their proportional share (1.03× and 0.81× per dimension), whereas the six joint angles draw 1.94× theirs—per dimension they are the most information-dense part of the descriptor, despite being the smallest block. (b) Within the angle block, the left ankle and left knee dominate, echoing the distal emphasis seen in the joint-level ranking.

TABLE XIII
POSITIONING AGAINST RELATED WORK. FIGURES ARE NOT DIRECTLY COMPARABLE: EACH ROW REPORTS A DIFFERENT TASK, CORPUS AND PROTOCOL

Method	**Venue**	**Task**	**Modality**	**Reported figure**	**Corpus / protocol**
Agarwal *et al.* [18]	CVPRW 2019	Deepfake det.	Facial + head motion	—	Per-individual profiles
FTCN [3]	ICCV 2021	Face forgery det.	Face RGB	~86–89% AUC	Celeb-DF, cross-dataset
RECCE [1]	CVPR 2022	Face forgery det.	Face RGB	~70.9% AUC	Celeb-DF-v2, cross-dataset
GaitFormer [12]	Sensors 2022	Gait recognition	Skeleton	92.5% acc.	CASIA-B
AltFreezing [4]	CVPR 2023	Face forgery det.	Face RGB	89.5% AUC	Celeb-DF, cross-dataset
GaitPT [11]	IEEE FG 2024	Gait recognition	Skeleton	82.6% acc.	CASIA-B
Petmezas *et al.* [56]	MTA 2024	Identity verification	Face + 3DMM	—	VoxCeleb2
TT-DF [22]	Preprint 2025	Body forgery det.	Pose / motion	—	TT-DF benchmark
This work	—	**Face-swap det.**	**Skeleton gait**	**94.95% AUC**	**GaitDeepfake-13, LOOCV**

murder case where the subject's face occupied roughly 10×5 pixels [57]. We cite this as an industry milestone and note its limits precisely—it is a single trade-press account of a forensic report admitted in a case not yet tried, not an appellate ruling or a peer-reviewed forensic validation, and the accompanying sub-0.1% EER is a vendor figure rather than an independent benchmark. It indicates institutional interest in gait as identity evidence; it does not establish gait biometrics as a validated forensic standard.

B. What This Approach Buys

Independence from synthesis quality. The detector never observes the manipulated region, so improvements in face-swap fidelity do not degrade it. This breaks the arms-race coupling that constrains artefact-based methods [1], [4]. The relevant adversarial question is not whether the face looks better but whether the body motion was modelled at all.

Uncorrelated failure modes. Because the evidence is disjoint from that used by facial detectors, the two families fail on different inputs. A fusion of the two is a natural and, we expect, productive combination—not because gait is stronger, but because it is independent.

Privacy characteristics. The pipeline reduces video to 12 skeletal coordinates per frame at the first stage. No facial imagery or face template is retained, and the stored enrolment signature is a 60×78 array of hip-centred kinematics. This is a materially smaller disclosure than a face template, though we note it is not anonymity: gait is itself a biometric, and an enrolment signature is personal data.

Degradation profile. Gait remains recoverable at distances and resolutions where facial analysis is not viable—the operative circumstance in the forensic case cited above [57].

C. Limitations and Threats to Validity

We list these in descending order of how much they should temper the results.

The ablation's negative result is specific to this scale. Every encoder configuration we tested is significantly worse than raw differencing (Section V-D), and we take that as justifying the deployed architecture. But 12 training subjects is a regime that penalises capacity, and the finding should not be read as a general claim that temporal encoders cannot help gait verification—only that, here, they reliably do not. Whether the ordering reverses on a CASIA-B-scale cohort is untested, and is the experiment we would run first given more data.

The face-swap validation is small. Section V-E now reports a measured outcome—3/3 clips correctly rejected, with similarity to the true body source above 0.9998 in every case—but $n = 3$ from a single generator under one masking configuration remains an existence check, not a statistically powered claim. It does not establish a detection rate with a confidence interval, and says nothing about other face-swap models, face-enhancement post-processing, or heavier recompression. Expanding to several dozen clips across at least two generators is the natural next step and is now the single most important gap.

Thirteen subjects is a small cohort. Augmentation multiplies sequences but not gaits; the effective sample for a subject-generalisation claim is 13. LOOCV at this size carries wide confidence intervals [46], which the ± 3.08 AUC and ± 6.82 recall standard deviations reflect. Validation on CASIA-B-scale cohorts is necessary before the numbers should be treated as population estimates. The cohort is also demographically narrow—university students aged roughly 19–25—so age-related and pathological gait variation is entirely unrepresented.

Conditions are controlled. Indoor recording, cooperative subjects, unobstructed full-body framing, two viewpoints. Occlusion, crowding, oblique and extreme viewing angles, carried loads and varied footwear are unevaluated, and each is known to affect gait recognition.

The threat model has a boundary. A generator that re-synthesises whole-body motion, or that retimes or warps the body, violates the assumption the method rests on. Such systems exist in the human-animation literature and are the subject of the emerging body-forgery benchmarks [21], [22]. Encouragingly, current generative animation appears not to preserve gait identity cues faithfully [24], which suggests gait may remain discriminative even against full-body synthesis—but that is a hypothesis, not a result of this work.

Scope of the verdict. The system adjudicates a claimed identity. It requires enrolment, and it cannot flag a synthetic person who claims no identity. It is a verification component, not a general-purpose deepfake detector.

Scores are not calibrated. As Fig. 8 shows, the output distribution is heavily saturated. The reported confidence should not be read as a probability without recalibration.

D. Future Work

The immediate priorities follow directly from the list above: correct and re-run the ablation; quantify detection on a substantially larger set of generated forgeries spanning multiple face-swap models; and extend the cohort in size and demographic range. Beyond these, graph convolutional modelling [10] suits the skeleton's natural topology better than the treatment of joints as a flat vector; per-subject threshold calibration addresses the precision–recall trade-off observed in Section V-B; and the disproportionate per-dimension value of the joint-angle block (Section VI) suggests a reduced descriptor suitable for embedded, real-time screening.

VIII. Conclusion

We have presented a face-swap deepfake detector that verifies identity from skeletal gait rather than from facial artefacts, exploiting the fact that face-region-only manipulation leaves the source body's locomotion intact. A difference-based temporal convolutional network, operating on 78-dimensional per-frame gait descriptors derived from 12 MediaPipe landmarks, attains a pooled ROC-AUC of 94.95%, accuracy of $87.04\% \pm 3.65\%$ and an equal error rate of $12.27\% \pm 3.80\%$ across 13 leave-one-subject-out folds and 2,240 verification pairs. Gradient attribution shows the decision resting on shoulder countermotion and distal foot kinematics, and identifies the six joint-angle dimensions as carrying nearly twice their proportional share of discriminative signal.

An ablation over 39 matched trials shows that the minimal difference network is not merely adequate but preferable: every encoder configuration we tested is significantly worse, and stripping the raw per-timestep comparison reduces the system to chance. We have also been explicit about what the experiments do not show. The face-swap validation, though it correctly rejected all three generated clips with similarity to the true body source above 0.9998 in every case, remains an existence check on a single generator rather than a statistically powered claim; thirteen subjects is a small basis for a generalisation claim; and the ablation's ordering is established at this cohort size, not in general. These are stated as they are so that the result which does stand—that skeletal gait discriminates claimed identity in subject-disjoint evaluation at 94.95% pooled AUC—can be read for what it is.

The broader argument is structural rather than numerical. Detectors that look for evidence inside the region an adversary controls are in a race they must keep re-entering. Looking instead at what the adversary declines to model changes the terms of the problem, and body motion is, for now, exactly that.

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